

Executive Summary

This report presents a comprehensive analysis of customer purchasing behavior, conducted to uncover the key trends shaping sales performance, customer engagement, and product demand. The workflow combined three complementary tools: Python for data cleaning and preparation, SQL for structured business querying, and Power BI for interactive visualization.

Across approximately 3,900 customer records, the analysis surfaces clear patterns in demographics, spending, subscription behavior, product preferences, and customer satisfaction — providing a foundation for confident, data-driven decisions on retention, marketing, and inventory strategy.

1. Project Objective

The analysis was guided by the following goals:

- Understand the core drivers of customer purchasing behavior.
- Identify key customer segments based on demographic attributes.
- Analyze spending patterns and subscription trends.
- Evaluate the relative performance of each product category.
- Surface factors that influence purchase decisions.
- Deliver an interactive dashboard to support stakeholder decision-making.

2. Data Collection & Preparation

Dataset Overview

The dataset captures customer transaction and behavioral details, including:

1. Customer ID	2. Age	3. Gender
4. Item Purchased	5. Category	6. Purchase Amount
7. Location	8. Size	9. Color
10. Season	11. Review Rating	12. Subscription Status
13. Shipping Type	14. Discount Applied	15. Promo Code Usage / Previous Purchases

Data Cleaning Process (Python)

Python and Pandas were used inside Jupyter Notebook to inspect, clean, and validate the raw dataset before it was loaded for analysis.

Activities performed:

- Imported the customer dataset and reviewed its structure and data types.
- Identified missing values — 37 blank entries were found in the Review Rating column.
- Imputed missing ratings using the category-wise median to preserve distribution integrity.
- Removed duplicate records and standardized column formats.
- Validated numerical and categorical fields ahead of database loading.

```
[1]: import pandas as pd;
df = pd.read_csv('customer_shopping_data.csv')
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	P
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	

```
[2]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Customer ID           3900 non-null   int64
1   Age                   3900 non-null   int64
2   Gender                3900 non-null   str
3   Item Purchased        3900 non-null   str
4   Category              3900 non-null   str
5   Purchase Amount (USD) 3900 non-null   int64
6   Location              3900 non-null   str
7   Size                  3900 non-null   str
8   Color                 3900 non-null   str
```

Figure 1. Initial dataset import and structure check (`df.head()`, `df.info()`) — 3,900 rows across 18 columns.

```

[3]: df.isnull().sum()

[3]: Customer ID      0
     Age             0
     Gender          0
     Item Purchased  0
     Category        0
     Purchase Amount (USD) 0
     Location        0
     Size            0
     Color           0
     Season          0
     Review Rating    37
     Subscription Status 0
     Shipping Type    0
     Discount Applied 0
     Promo Code Used  0
     Previous Purchases 0
     Payment Method   0
     Frequency of Purchases 0
     dtype: int64

[5]: df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))

[6]: df.isnull().sum()

[6]: Customer ID      0
     Age             0
     Gender          0
     Item Purchased  0
     Category        0
     Purchase Amount (USD) 0
     Location        0
     Size            0
     Color           0
     Season          0
     Review Rating    0
     Subscription Status 0
  
```

Figure 2. Missing-value audit and median imputation of the Review Rating field.

Outcome:

A clean, structured, and validated dataset — ready for reliable downstream analysis.

3. Database Integration

The cleaned dataset was loaded into a MySQL database to enable structured, repeatable business querying.

Activities performed:

- Connected to MySQL via SQLAlchemy using a Python engine.
- Pushed the cleaned dataframe into a customer table within the customer_behavior database.
- Verified the load with a direct SQL read-back of the table.

The screenshot shows a Jupyter Notebook interface with a Python 3 (ipykernel) environment. The code in the cell [7]:

```
[7]: from sqlalchemy import create_engine
username = "root"
password = "password"
host = "localhost"
port = "3306"
database = "customer_behavior"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}")

table_name = "customer" # choose any table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

pd.read_sql("SELECT * FROM customer LIMIT 5;", engine)
```

The output of the code is a table with 18 columns: Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Location, Size, Color, Season, Review Rating, Subscription Status, Shipping Type, Discount Applied, Promo Code Used, Previous Purchases, and P. The table displays 5 rows of data:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	P
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	
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Figure 3. SQLAlchemy connection loading the cleaned dataset into the MySQL customer_behavior database.

4. SQL-Based Business Analysis

With the data in MySQL, SQL queries were used to answer specific business questions across customer demographics, spending, and product performance.

Revenue by Gender

Aggregating total purchase amount by gender highlights a clear skew in revenue contribution between segments.

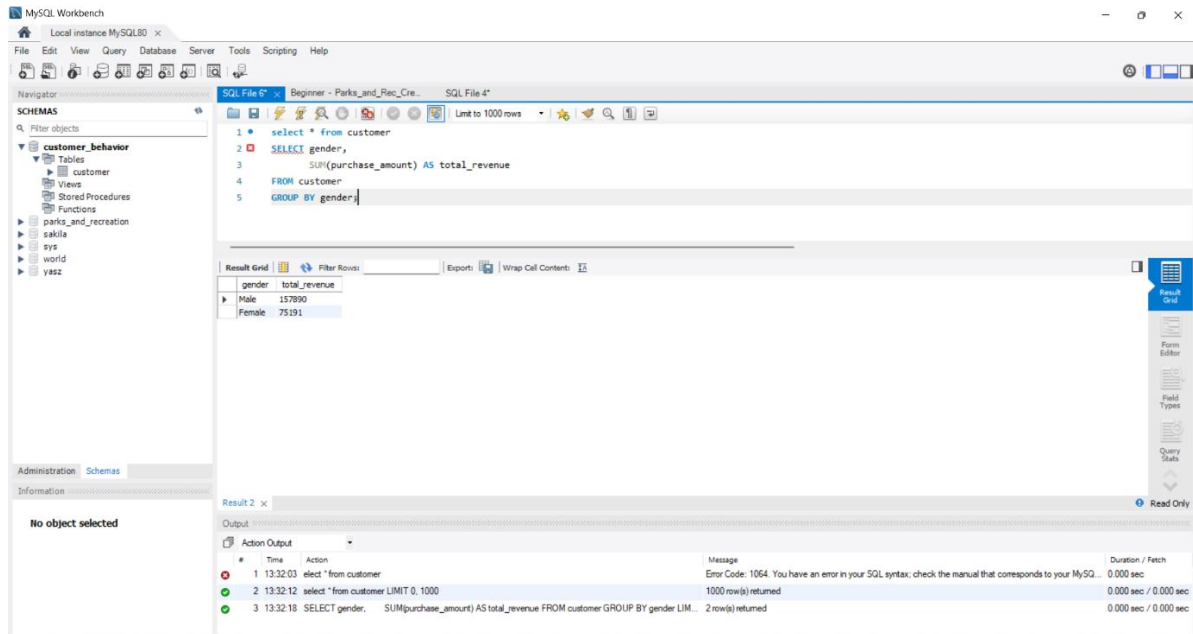


Figure 4. Total revenue grouped by customer gender.

Subscription Status vs. Spend

Grouping customers by subscription status compares customer counts, average spend, and total revenue between subscribed and non-subscribed segments.

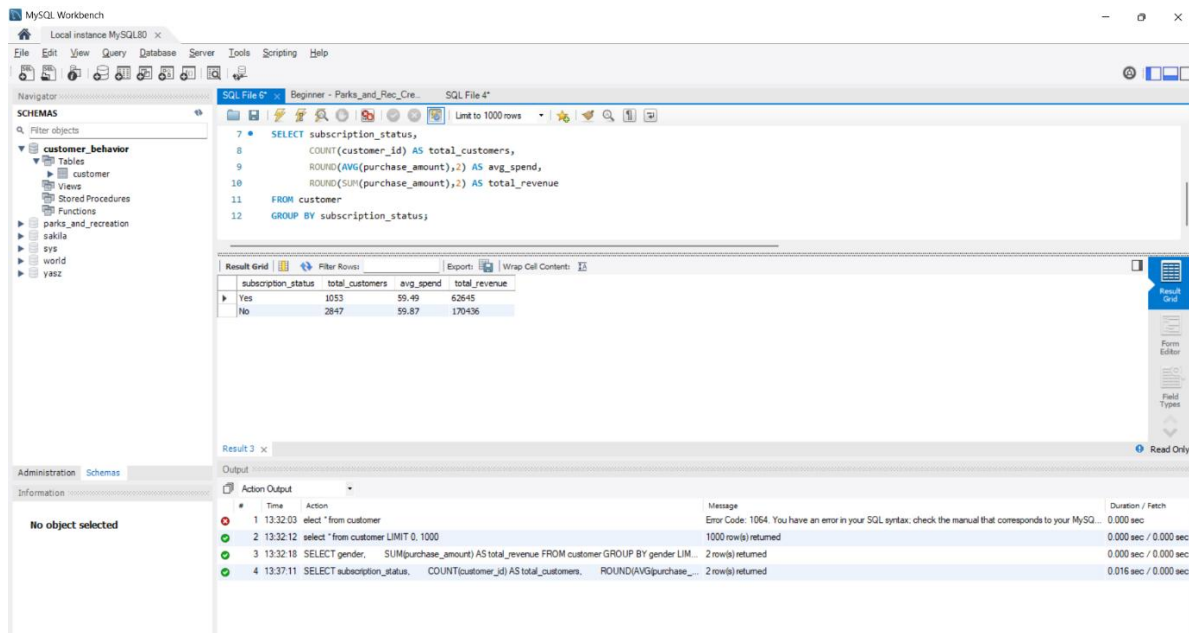


Figure 5. Customer count, average spend, and total revenue by subscription status.

Top-Selling Items by Category

A window function (ROW_NUMBER with PARTITION BY) was used to rank items within each category by order volume, surfacing the best-selling item per category.

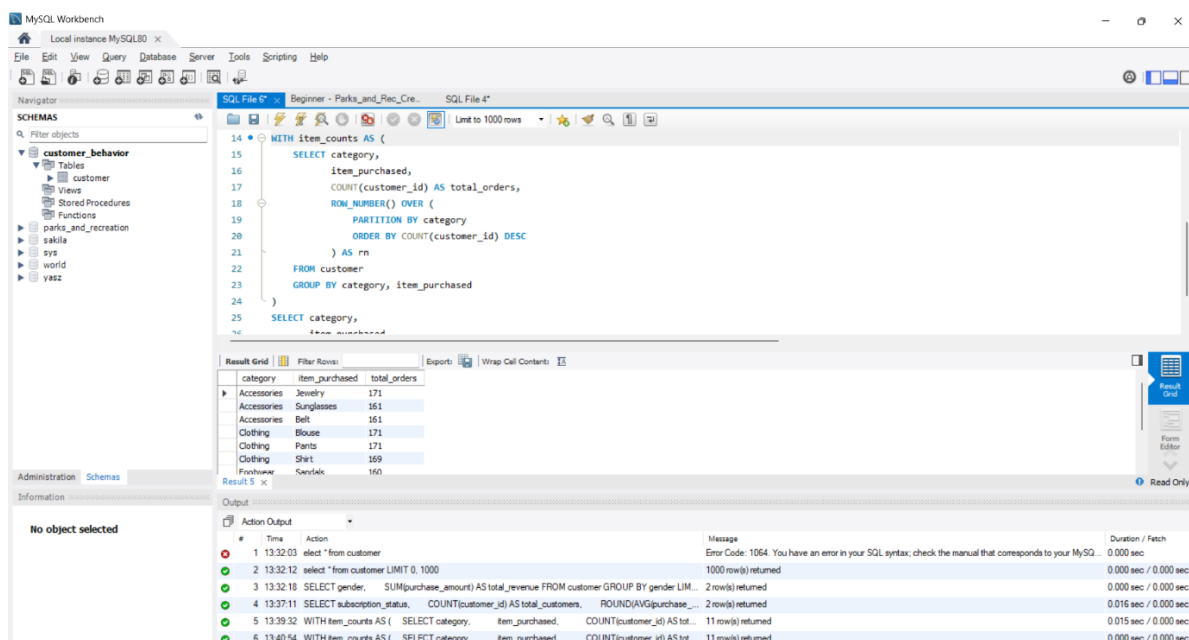


Figure 6. Top-ranked item per product category by total orders.

5. Dashboard Development (Power BI)

An interactive Power BI dashboard was built to bring the SQL and Python analysis together into a single, stakeholder-facing view of customer behavior.

Key Performance Indicators

3.9K Total Customers	\$59.76 Average Purchase Amount	3.75 Average Review Rating
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Dashboard Visualizations

- Subscription Status Analysis — distribution of customers with and without an active subscription.
- Purchase Amount by Category — the product categories generating the highest revenue.
- Purchase Trends by Season — seasonal shifts in customer demand.
- Customer Distribution by Age Group — concentration of customers across age ranges.
- Review Rating by Category — customer satisfaction compared across categories.

Interactive Filters

The dashboard includes slicers for Age Group, Subscription Status, Gender, Category, Season, and Location — allowing stakeholders to drill into specific customer segments dynamically.

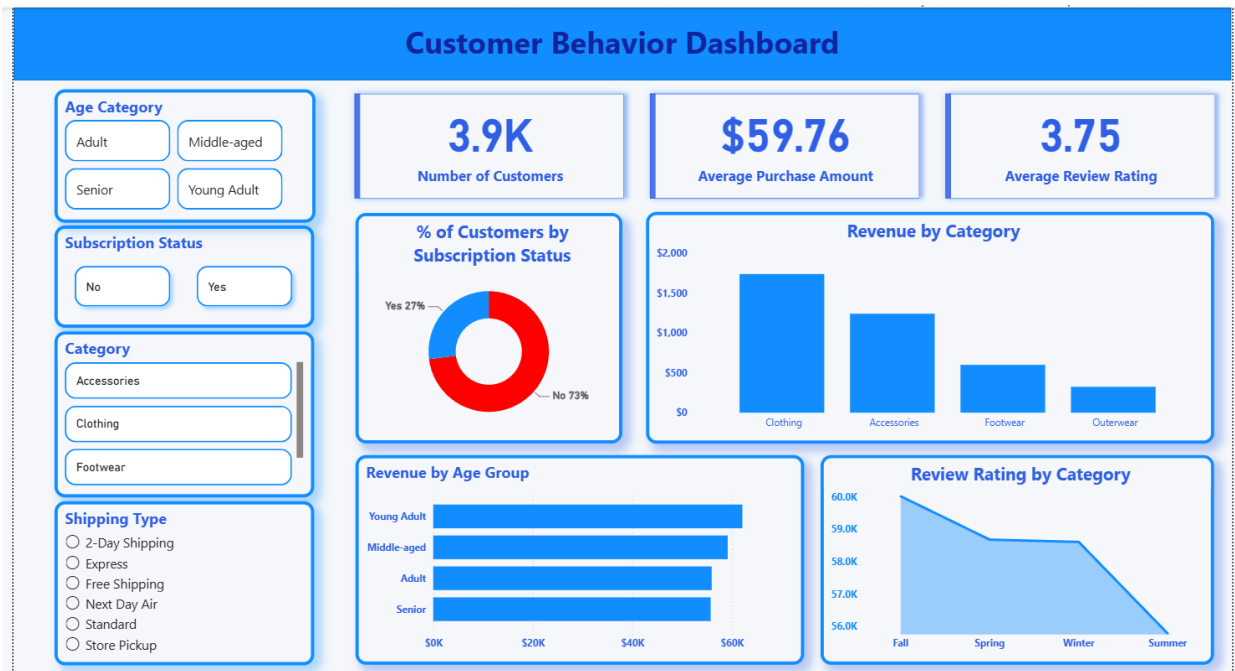


Figure 7. Power BI Customer Behavior Dashboard — KPIs, segmentation filters, and category/season breakdowns.

6. Key Insights

- The customer base totals approximately 3,900 customers.
- Average customer spending is just under \$60 per purchase.
- Customer review ratings reflect moderate-to-high overall satisfaction.
- A small set of product categories drive a disproportionate share of revenue.
- Subscription status is associated with measurable differences in purchasing behavior and engagement.
- Seasonal patterns meaningfully affect purchasing volume and timing.
- Age demographics reveal distinct segments with different spending profiles.

7. Business Recommendations

- Prioritize investment and merchandising around the highest-performing product categories.
- Strengthen subscription programs to improve long-term customer retention.
- Design targeted marketing campaigns tailored to specific age segments.
- Use seasonal demand trends to guide inventory planning and promotional timing.
- Address lower-rated categories to lift overall customer satisfaction.
- Expand discount and loyalty incentives to encourage repeat purchases.

8. Conclusion

This Customer Behavior Analysis project turned raw transaction data into clear, actionable business insight. By combining Python-based data cleaning, SQL-driven querying, and Power BI visualization, the project gives stakeholders a complete view of customer purchasing patterns and overall business performance.

The resulting dashboard supports ongoing monitoring of key metrics and provides a practical foundation for decisions on customer retention, sales growth, and business optimization going forward.